

Interpolative Separable Density Fitting and Tensor Hypercontraction of Two-Electron Integrals for Ultrasoft and Projector-Augmented-Wave Pseudopotentials

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Interpolative separable density fitting (ISDF) provides a tensor-hypercontracted (THC) representation of two-electron repulsion integrals that has enabled low-scaling random phase approximation (RPA) and *GW* calculations for crystalline systems, but existing momentum-resolved formulations are restricted to norm-conserving pseudopotentials. We extend ISDF and THC to the two augmented pseudopotential families that dominate plane-wave electronic structure—ultrasoft (USPP) and projector-augmented-wave (PAW)—by applying ISDF to the augmented (compensated) pair-density algebra rather than to the smooth pseudo pair density alone. The resulting factorization

$$V_{ijkl}^{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l} \approx \sum_{\Lambda \Sigma} X_{\Lambda i}^{\mathbf{k}_i *} X_{\Lambda j}^{\mathbf{k}_j} \mathcal{V}_{\Lambda \Sigma}^{\mathbf{q}} X_{\Sigma k}^{\mathbf{k}_k *} X_{\Sigma l}^{\mathbf{k}_l}$$

retains the standard THC structure, with a composite auxiliary index Λ that combines smooth ISDF interpolation rows with atom-centered augmentation rows, and a projected Coulomb matrix $\mathcal{V}^{\mathbf{q}}$ equal to the Coulomb metric of the compensated auxiliary functions plus a same-atom one-center correction. Ultrasoft and PAW are treated on the same footing: the augmentation charges enter the composite basis in both cases, while the one-center all-electron residual—built from PAW partial waves—is present only for PAW, so that the ultrasoft scheme is recovered as the special case in which it is absent. The construction preserves the separability in orbital and momentum indices and the linear-in- N_k structure required for low-scaling many-body perturbation theory. We give the complete density decomposition, compensation-charge construction, local channel factorization, kernel assembly, error estimates in the Coulomb metric, and numerical demonstrations: the compressed integrals converge systematically to the reference ultrasoft and PAW integrals for LiH, and the silicon RPA correlation energy evaluated entirely within the factorization agrees, for both ultrasoft and PAW, with norm-conserving references to within the spread between norm-conserving potentials.

I. INTRODUCTION

Two-electron repulsion integrals (ERIs) over all-electron single-particle orbitals are central to Hartree-Fock theory, hybrid density functionals, random phase approximation (RPA), *GW*, vertex-corrected Green’s-function methods, quantum embedding, and wave-function-based correlation theories. For crystalline systems, a generic Bloch-basis ERI is a rank-four tensor with orbital and momentum labels,

$$V_{ijkl}^{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l} = \int_{\Omega} d\mathbf{r} \int_{\Omega} d\mathbf{r}' \frac{1}{|\mathbf{r} - \mathbf{r}'|} \times \varphi_i^{\mathbf{k}_i *}(\mathbf{r}) \varphi_j^{\mathbf{k}_j}(\mathbf{r}) \varphi_k^{\mathbf{k}_k *}(\mathbf{r}') \varphi_l^{\mathbf{k}_l}(\mathbf{r}'). \quad (1)$$

subject to crystal-momentum conservation. In a direct representation, storage and contractions involving Eq. (1) are prohibitive. Tensor hypercontraction (THC) overcomes this by expressing the ERI as a product of five second-order tensors, fully separating the four orbital indices. Interpolative separable density fitting (ISDF) is a systematic route to THC: it compresses the set of pair densities by selecting interpolation points and auxiliary functions, leading directly to a compact projected Coulomb matrix.

Existing momentum-resolved ISDF-THC formula-

tions [11, 12] were developed for norm-conserving pseudopotentials, whose pair densities are smooth and interpolate well on moderate uniform grids. The two pseudopotential families that dominate plane-wave electronic structure, however—ultrasoft pseudopotentials (USPP) [3] and the projector-augmented-wave (PAW) method [1, 2], as implemented in widely used codes such as VASP, Quantum ESPRESSO, and ABINIT—introduce localized augmentation that a smooth grid does not capture. In USPP the physical charge density is the smooth pseudo density plus atom-centered augmentation charges that restore the correct multipoles; PAW goes further, reconstructing the all-electron wave function from smooth pseudo orbitals and atom-centered all-electron partial waves whose near-nucleus nodal structure is essential for accurate Coulomb matrix elements. In either case, resolving the augmentation directly on a uniform grid would destroy the compactness that makes ISDF attractive. The objective of this paper is to extend ISDF and THC to both schemes in a way that preserves their advantages: smooth-grid ISDF in the interstitial region and compact atom-local augmentation inside the spheres.

The key idea is simple but important: ISDF should not be applied to the smooth pseudo pair density and the result reinterpreted as the augmented (or all-electron)

factorization. Instead, one first rewrites the augmented Coulomb integral in terms of compensated smooth pair densities plus strictly one-center corrections. ISDF is then applied to the compensated smooth part, while a separate low-rank factorization handles the finite-dimensional one-center correction. Ultrasoft and PAW differ only in this one-center block: it vanishes for USPP, whose augmentation enters solely through the compensated smooth density, and it is the all-electron partial-wave residual for PAW. The result is a THC factorization with the same structure as the momentum-dependent ISDF representation of Yeh and Morales,

$$V_{ijkl}^{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l} \approx \sum_{\mu\nu} X_{\mu i}^{\mathbf{k}_i*} X_{\mu j}^{\mathbf{k}_j} V_{\mu\nu}^{\mathbf{q}} X_{\nu k}^{\mathbf{k}_k*} X_{\nu l}^{\mathbf{k}_l}, \quad (2)$$

but with an enlarged auxiliary index containing the augmentation features. The notation below follows Refs. 11 and 12: $\varphi_i^{\mathbf{k}}$ denotes a Bloch orbital, $X_{\mu i}^{\mathbf{k}}$ the collocation matrix, $\zeta_{\mu}^{\mathbf{q}}$ the ISDF auxiliary function at transferred momentum $\mathbf{q}$, $V_{\mu\nu}^{\mathbf{q}}$ the projected Coulomb matrix, N_{μ} the number of interpolation points, and $\alpha = N_{\mu}/N_{\text{orb}}$ the ISDF compression ratio.

II. MOMENTUM-RESOLVED ISDF AND THC NOTATION

Consider Bloch orbitals

$$\varphi_i^{\mathbf{k}}(\mathbf{r}) = u_i^{\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}, \quad (3)$$

where $u_i^{\mathbf{k}}$ is lattice-periodic. With the momentum-transfer convention used in Refs. 11 and 12, the ERI in Eq. (1) is nonzero when

$$\mathbf{q} = \mathbf{k}_i - \mathbf{k}_j + \mathbf{G} = \mathbf{k}_l - \mathbf{k}_k + \mathbf{G}' \quad (4)$$

for reciprocal lattice vectors $\mathbf{G}$ and $\mathbf{G}'$. The first pair density has crystal momentum $-\mathbf{q}$, while the second has crystal momentum $+\mathbf{q}$. For an arbitrary fixed $\mathbf{q}$, define the family of pair densities

$$\rho^{\mathbf{q}}(\mathbf{k}ij, \mathbf{r}) = \varphi_i^{\mathbf{k}-\mathbf{q}*}(\mathbf{r})\varphi_j^{\mathbf{k}}(\mathbf{r}). \quad (5)$$

ISDF approximates Eq. (5) as

$$\rho^{\mathbf{q}}(\mathbf{k}ij, \mathbf{r}) \approx \sum_{\mu=1}^{N_{\mu}} \varphi_i^{\mathbf{k}-\mathbf{q}*}(\mathbf{r}_{\mu})\varphi_j^{\mathbf{k}}(\mathbf{r}_{\mu})\zeta_{\mu}^{\mathbf{q}}(\mathbf{r}), \quad (6)$$

where $\{\mathbf{r}_{\mu}\}$ are interpolating points and $\{\zeta_{\mu}^{\mathbf{q}}\}$ are auxiliary functions. Equivalently,

$$\rho^{\mathbf{q}}(\mathbf{k}ij, \mathbf{r}) \approx \sum_{\mu} X_{\mu i}^{\mathbf{k}-\mathbf{q}*} X_{\mu j}^{\mathbf{k}} \zeta_{\mu}^{\mathbf{q}}(\mathbf{r}), \quad X_{\mu i}^{\mathbf{k}} = \varphi_i^{\mathbf{k}}(\mathbf{r}_{\mu}). \quad (7)$$

The corresponding projected Coulomb matrix is

$$V_{\mu\nu}^{\mathbf{q}} = \int_{\Omega} d\mathbf{r} \int_{\Omega} d\mathbf{r}' \zeta_{\mu}^{-\mathbf{q}}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \zeta_{\nu}^{\mathbf{q}}(\mathbf{r}'), \quad (8)$$

with the usual periodic treatment of the Coulomb singularity. Inserting Eq. (7) into Eq. (1) gives

$$V_{ijkl}^{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l} \approx \sum_{\mu\nu} X_{\mu i}^{\mathbf{k}_i*} X_{\mu j}^{\mathbf{k}_j} V_{\mu\nu}^{\mathbf{q}} X_{\nu k}^{\mathbf{k}_k*} X_{\nu l}^{\mathbf{k}_l}. \quad (9)$$

After the interpolation points are selected, the auxiliary functions may be obtained from a least-squares problem. Written in the notation of Ref. 11, let the pair-density matrix be $\rho^{\mathbf{q}}(\mathbf{k}ij, \mathbf{r})$. The overdetermined system is

$$C^{\mathbf{q}}\Theta^{\mathbf{q}} = Z^{\mathbf{q}}, \quad (10)$$

with

$$Z_{\nu r}^{\mathbf{q}} = \sum_{\mathbf{k}ij} \rho^{\mathbf{q}*}(\mathbf{k}ij, \mathbf{r}_{\nu}) \rho^{\mathbf{q}}(\mathbf{k}ij, \mathbf{r}), \quad (11a)$$

$$C_{\nu\mu}^{\mathbf{q}} = \sum_{\mathbf{k}ij} \rho^{\mathbf{q}*}(\mathbf{k}ij, \mathbf{r}_{\nu}) \rho^{\mathbf{q}}(\mathbf{k}ij, \mathbf{r}_{\mu}), \quad (11b)$$

$$\Theta_{\mu r}^{\mathbf{q}} = \zeta_{\mu}^{\mathbf{q}}(\mathbf{r}). \quad (11c)$$

A common point-selection strategy is pivoted Cholesky on

$$S^{\mathbf{q}} = (\rho^{\mathbf{q}})^{\dagger} \rho^{\mathbf{q}}, \quad (12)$$

or the equivalent column-pivoted QR factorization of $\rho^{\mathbf{q}}$. Empirically, $N_{\mu} = \alpha N_{\text{orb}}$ with moderate α is often sufficient for chemically useful accuracy in pseudopotential periodic calculations.

III. PAW ALL-ELECTRON PAIR DENSITIES

A. PAW transformation

The PAW transformation maps smooth pseudo orbitals to all-electron valence orbitals. For a Bloch orbital, we write

$$|\varphi_i^{\mathbf{k}}\rangle = |\tilde{\varphi}_i^{\mathbf{k}}\rangle + \sum_a \sum_{\alpha} \left(|\phi_{a\alpha}^{\mathbf{k}}\rangle - |\tilde{\phi}_{a\alpha}^{\mathbf{k}}\rangle \right) P_{i,a\alpha}^{\mathbf{k}}, \quad (13)$$

where a labels atoms in the primitive cell, α labels partial-wave channels on atom a , and

$$P_{i,a\alpha}^{\mathbf{k}} = \langle \tilde{p}_{a\alpha}^{\mathbf{k}} | \tilde{\varphi}_i^{\mathbf{k}} \rangle \quad (14)$$

is the PAW projector overlap. The Bloch-summed projector is

$$\tilde{p}_{a\alpha}^{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\tau}_a)} \tilde{p}_{a\alpha}(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_a), \quad (15)$$

up to the normalization convention used for Bloch functions. Within each augmentation sphere, the pseudo

orbital and the all-electron orbital have simultaneous partial-wave expansions,

$$\varphi_i^{\mathbf{k}}(\mathbf{r}) \simeq \sum_{\alpha} \phi_{a\alpha}(\mathbf{r} - \boldsymbol{\tau}_a) P_{i,a\alpha}^{\mathbf{k}}, \quad (16a)$$

$$\tilde{\varphi}_i^{\mathbf{k}}(\mathbf{r}) \simeq \sum_{\alpha} \tilde{\phi}_{a\alpha}(\mathbf{r} - \boldsymbol{\tau}_a) P_{i,a\alpha}^{\mathbf{k}}. \quad (16b)$$

The equalities become exact in the complete partial-wave limit. In the remainder, frozen core contributions are suppressed unless explicitly stated; the target ERI tensor is the valence all-electron tensor reconstructed by PAW.

B. Uncompensated PAW pair-density decomposition

For the pair $(i\mathbf{k}_i, j\mathbf{k}_j)$, define

$$\rho_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \varphi_i^{\mathbf{k}_i*}(\mathbf{r})\varphi_j^{\mathbf{k}_j}(\mathbf{r}), \quad \tilde{\rho}_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \tilde{\varphi}_i^{\mathbf{k}_i*}(\mathbf{r})\tilde{\varphi}_j^{\mathbf{k}_j}(\mathbf{r}). \quad (17)$$

The one-center all-electron and pseudo pair densities are

$$\rho_{ij}^{1,a;\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \sum_{\alpha\beta} P_{i,a\alpha}^{\mathbf{k}_i*} P_{j,a\beta}^{\mathbf{k}_j} \phi_{a\alpha}^*(\mathbf{r} - \boldsymbol{\tau}_a) \phi_{a\beta}(\mathbf{r} - \boldsymbol{\tau}_a), \quad (18a)$$

$$\tilde{\rho}_{ij}^{1,a;\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \sum_{\alpha\beta} P_{i,a\alpha}^{\mathbf{k}_i*} P_{j,a\beta}^{\mathbf{k}_j} \tilde{\phi}_{a\alpha}^*(\mathbf{r} - \boldsymbol{\tau}_a) \tilde{\phi}_{a\beta}(\mathbf{r} - \boldsymbol{\tau}_a). \quad (18b)$$

Then the all-electron PAW pair density has the standard additive form

$$\rho_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \tilde{\rho}_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) + \sum_a \left[\rho_{ij}^{1,a;\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) - \tilde{\rho}_{ij}^{1,a;\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) \right]. \quad (19)$$

This expression alone is not a useful way to compute ERIs, because the Coulomb interaction couples localized

augmentation corrections on different atoms and couples localized radial-grid quantities to smooth-grid quantities.

C. Compensation charges and PAW Coulomb partition

Define the channel product difference

$$\Delta Q_{a,\alpha\beta}(\mathbf{s}) = \phi_{a\alpha}^*(\mathbf{s})\phi_{a\beta}(\mathbf{s}) - \tilde{\phi}_{a\alpha}^*(\mathbf{s})\tilde{\phi}_{a\beta}(\mathbf{s}), \quad \mathbf{s} = \mathbf{r} - \boldsymbol{\tau}_a. \quad (20)$$

For every pair of partial-wave channels, introduce a smooth localized compensation charge $\hat{Q}_{a,\alpha\beta}(\mathbf{s})$ such that

$$\int_{\Omega_a} d\mathbf{s} |\mathbf{s}|^\ell Y_{\ell m}(\hat{\mathbf{s}}) \left[\Delta Q_{a,\alpha\beta}(\mathbf{s}) - \hat{Q}_{a,\alpha\beta}(\mathbf{s}) \right] = 0 \quad (21)$$

for all included multipoles (ℓm). Equivalently, $\Delta Q_{a,\alpha\beta} - \hat{Q}_{a,\alpha\beta}$ has zero multipole moments through the PAW compensation order, so its electrostatic potential vanishes outside the augmentation sphere.

The pair-specific compensation charge is

$$\hat{\rho}_{ij}^{a;\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \sum_{\alpha\beta} P_{i,a\alpha}^{\mathbf{k}_i*} P_{j,a\beta}^{\mathbf{k}_j} \hat{Q}_{a,\alpha\beta}(\mathbf{r} - \boldsymbol{\tau}_a), \quad (22)$$

and its Bloch-periodic form at momentum $\mathbf{k}_j - \mathbf{k}_i$ is obtained by the corresponding lattice sum. The compensated smooth pair density is

$$\bar{\rho}_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \tilde{\rho}_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) + \sum_a \hat{\rho}_{ij}^{a;\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}). \quad (23)$$

Using the zero-multipole property, the all-electron PAW Coulomb integral can be partitioned into a smooth compensated part and one-center residual corrections,

$$V_{ijkl}^{\mathbf{k}_i\mathbf{k}_j\mathbf{k}_k\mathbf{k}_l,\text{PAW}} = \left(\bar{\rho}_{ij}^{\mathbf{k}_i\mathbf{k}_j} \middle| \bar{\rho}_{kl}^{\mathbf{k}_k\mathbf{k}_l} \right) + \sum_a \Delta V_{ij,kl}^{a;\mathbf{k}_i\mathbf{k}_j\mathbf{k}_k\mathbf{k}_l}, \quad (24)$$

$$\Delta V_{ij,kl}^{a;\mathbf{k}_i\mathbf{k}_j\mathbf{k}_k\mathbf{k}_l} = \sum_{\alpha\beta\gamma\delta} P_{i,a\alpha}^{\mathbf{k}_i*} P_{j,a\beta}^{\mathbf{k}_j} \Delta C_{a,\alpha\beta,\gamma\delta} P_{k,a\gamma}^{\mathbf{k}_k*} P_{l,a\delta}^{\mathbf{k}_l}, \quad (25)$$

$$\Delta C_{a,\alpha\beta,\gamma\delta} = \left(\phi_{a\alpha}^* \phi_{a\beta} \middle| \phi_{a\gamma}^* \phi_{a\delta} \right)_a - \left(\tilde{\phi}_{a\alpha}^* \tilde{\phi}_{a\beta} + \hat{Q}_{a,\alpha\beta} \middle| \tilde{\phi}_{a\gamma}^* \tilde{\phi}_{a\delta} + \hat{Q}_{a,\gamma\delta} \right)_a. \quad (26)$$

Here $(f|g)$ denotes the Coulomb bilinear form,

$$(f|g) = \int d\mathbf{r} \int d\mathbf{r}' f(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} g(\mathbf{r}'), \quad (27)$$

and the subscript a means that the one-center radial/angular representation of atom a is used. Equations

(24) is the PAW object to compress. It is exact in the complete partial-wave and compensation-multipole limits and avoids long-range cross terms between augmentation spheres.

IV. PAW-ISDF ANSATZ

A. Composite auxiliary index

The PAW-ISDF construction uses a composite auxiliary index

$$\Lambda \in \mathcal{G} \cup \mathcal{A}, \quad (28)$$

where $\mathcal{G}$ labels global smooth ISDF interpolation rows and $\mathcal{A}$ labels atom-centered augmentation rows. The goal is to interpolate the compensated smooth pair density in Eq. (23) as

$$\bar{\rho}_{ij}^{\mathbf{k}_i \mathbf{k}_j}(\mathbf{r}) \approx \sum_{\Lambda} X_{\Lambda i}^{\mathbf{k}_i *} X_{\Lambda j}^{\mathbf{k}_j} \zeta_{\Lambda}^{\mathbf{k}_j - \mathbf{k}_i}(\mathbf{r}). \quad (29)$$

For the ERI convention in Eq. (4), the first pair uses $\zeta_{\Lambda}^{-\mathbf{q}}$ and the second pair uses $\zeta_{\Sigma}^{\mathbf{q}}$.

For global rows $\Lambda = \mu \in \mathcal{G}$,

$$\begin{aligned} X_{\mu i}^{\mathbf{k}} &= \tilde{\varphi}_i^{\mathbf{k}}(\mathbf{r}_{\mu}), \\ \zeta_{\mu}^{\mathbf{q}}(\mathbf{r}) &= \text{smooth ISDF auxiliary function.} \end{aligned} \quad (30)$$

These rows interpolate the smooth pseudo pair density.

For atom-centered rows $\Lambda = (a\lambda) \in \mathcal{A}$, define a local channel factor $U_{a,\lambda\alpha}$ and the projector-derived orbital feature

$$Y_{a\lambda,i}^{\mathbf{k}} \equiv X_{a\lambda,i}^{\mathbf{k}} = \sum_{\alpha} U_{a,\lambda\alpha} P_{i,a\alpha}^{\mathbf{k}}. \quad (31)$$

The associated auxiliary function is a Bloch sum of an atom-local function,

$$\eta_{a\lambda}^{\mathbf{q}}(\mathbf{r}) = \sum_{\mathbf{R}} e^{i\mathbf{q} \cdot (\mathbf{R} + \boldsymbol{\tau}_a)} \eta_{a\lambda}(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_a), \quad (32)$$

and

$$\zeta_{a\lambda}^{\mathbf{q}}(\mathbf{r}) = \eta_{a\lambda}^{\mathbf{q}}(\mathbf{r}). \quad (33)$$

The local ISDF condition for the compensation-charge products is

$$\hat{Q}_{a,\alpha\beta}(\mathbf{s}) \approx \sum_{\lambda} U_{a,\lambda\alpha}^* U_{a,\lambda\beta} \eta_{a\lambda}(\mathbf{s}). \quad (34)$$

Inserting Eq. (34) into Eq. (22) gives

$$\hat{\rho}_{ij}^{a;\mathbf{k}_i \mathbf{k}_j}(\mathbf{r}) \approx \sum_{\lambda} Y_{a\lambda,i}^{\mathbf{k}_i *} Y_{a\lambda,j}^{\mathbf{k}_j} \eta_{a\lambda}^{\mathbf{k}_j - \mathbf{k}_i}(\mathbf{r}). \quad (35)$$

Thus Eq. (29) has the same separable orbital structure as ordinary ISDF, but with additional rows based on PAW

projector overlaps rather than real-space pseudo-orbital values.

B. Projected Coulomb matrix for compensated densities

Define the compensated projected Coulomb matrix

$$V_{\Lambda\Sigma}^{\mathbf{q},C} = \int_{\Omega} d\mathbf{r} \int_{\Omega} d\mathbf{r}' \zeta_{\Lambda}^{-\mathbf{q}}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \zeta_{\Sigma}^{\mathbf{q}}(\mathbf{r}'). \quad (36)$$

In reciprocal space, for a neutral periodic pair-density block, this can be evaluated as

$$V_{\Lambda\Sigma}^{\mathbf{q},C} = \frac{4\pi}{\Omega} \sum_{\mathbf{G}}' \frac{\zeta_{\Lambda}^{-\mathbf{q}}(\mathbf{G}) \zeta_{\Sigma}^{\mathbf{q}}(-\mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2}, \quad (37)$$

where the prime indicates the standard treatment of the $\mathbf{q} + \mathbf{G} = 0$ Coulomb singularity. In practice, the smooth-smooth block is computed with FFT or Poisson solves; the smooth-local and local-local blocks can be computed by FFT after depositing the local functions on the smooth grid, by reciprocal-space analytic transforms of localized functions, or by multipole/radial-grid integration.

C. Local PAW residual in the same THC space

The one-center correction tensor ΔC_a in Eq. (26) is finite-dimensional. We factor it in the same local channel basis used for the compensation functions,

$$\Delta C_{a,\alpha\beta,\gamma\delta} \approx \sum_{\lambda\xi} U_{a,\lambda\alpha}^* U_{a,\lambda\beta} K_{a,\lambda\xi} U_{a,\xi\gamma}^* U_{a,\xi\delta}. \quad (38)$$

This equation can be viewed as a local THC factorization in PAW channel-product space. It is exact if the local rank includes all independent channel products; otherwise K_a is obtained by a Coulomb-metric or Frobenius-metric least-squares fit. Inserting Eq. (38) into Eq. (25) yields

$$\Delta V_{ij,kl}^{a;\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l} \approx \sum_{\lambda\xi} Y_{a\lambda,i}^{\mathbf{k}_i *} Y_{a\lambda,j}^{\mathbf{k}_j} K_{a,\lambda\xi} Y_{a\xi,k}^{\mathbf{k}_k *} Y_{a\xi,l}^{\mathbf{k}_l}. \quad (39)$$

V. FINAL PAW-ISDF-THC ERI FORMULA

Combining the compensated Coulomb term and the one-center residual correction gives the proposed PAW-ISDF-THC representation:

$$V_{ijkl}^{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l, \text{PAW-ISDF}} = \sum_{\Lambda \Sigma} X_{\Lambda i}^{\mathbf{k}_i *} X_{\Lambda j}^{\mathbf{k}_j} \mathcal{V}_{\Lambda \Sigma}^{\mathbf{q}} X_{\Sigma k}^{\mathbf{k}_k *} X_{\Sigma l}^{\mathbf{k}_l} \quad (40)$$

where

$$\mathcal{V}_{\Lambda \Sigma}^{\mathbf{q}} = V_{\Lambda \Sigma}^{\mathbf{q}, C} + \sum_a \delta_{\Lambda, a\lambda} \delta_{\Sigma, a\xi} K_{a, \lambda\xi} \quad (41)$$

and

$$X_{\Lambda i}^{\mathbf{k}} = \begin{cases} \tilde{\varphi}_i^{\mathbf{k}}(\mathbf{r}_\mu), & \Lambda = \mu \in \mathcal{G}, \\ \sum_\alpha U_{a, \lambda\alpha} P_{i, a\alpha}^{\mathbf{k}}, & \Lambda = (a\lambda) \in \mathcal{A}. \end{cases} \quad (42)$$

The first term in Eq. (41) is the Coulomb metric of the compensated ISDF auxiliary functions, while the second term restores the all-electron one-center PAW correction that was subtracted when the compensated smooth density was introduced. Equation (40) is the desired THC form for all-electron PAW ERIs.

Hermiticity follows from the properties

$$[V_{\Lambda \Sigma}^{\mathbf{q}, C}]^* = V_{\Sigma \Lambda}^{-\mathbf{q}, C}, \quad K_{a, \lambda\xi}^* = K_{a, \xi\lambda}, \quad (43)$$

which imply

$$(V_{ijkl}^{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l})^* = V_{jilk}^{\mathbf{k}_j \mathbf{k}_i \mathbf{k}_l \mathbf{k}_k}. \quad (44)$$

A. Ultrasoft and norm-conserving pseudopotentials as special cases

The construction above specializes immediately to ultrasoft pseudopotentials, which is what places USPP and PAW within a single factorization. In USPP the augmented charge density is the smooth pseudo density plus atom-centered augmentation charges $Q_{a, \alpha\beta}(\mathbf{r})$ [3] that restore the multipoles of the all-electron-minus-pseudo density difference—exactly the role played by the compensation charges $\hat{Q}_{a, \alpha\beta}$ of Eq. (21). Unlike PAW, however, the ultrasoft scheme does not reconstruct the all-electron wave function inside the sphere, so it carries no one-center all-electron Coulomb residual: $\Delta C_a = 0$, and therefore $K_a = 0$ in Eq. (38). The factorization is then *identical* to Eqs. (40)–(42), with the same composite collocation matrix X , but with the projected Coulomb matrix reduced to its compensated part,

$$\mathcal{V}_{\Lambda \Sigma}^{\mathbf{q}}|_{\text{USPP}} = V_{\Lambda \Sigma}^{\mathbf{q}, C}, \quad (45)$$

i.e. the same-atom K_a block is simply absent. Norm-conserving pseudopotentials are the further special case in which the augmentation rows are empty ($\mathcal{A} = \emptyset$) and Λ runs over the smooth ISDF points only, recovering the standard momentum-resolved ISDF-THC of Refs. 11 and

12. The three pseudopotential families thus form a hierarchy within one factorization, distinguished only by which parts of the composite auxiliary index and projected Coulomb matrix are populated: norm-conserving (smooth rows only), ultrasoft (smooth and augmentation rows, no K_a), and PAW (smooth and augmentation rows plus the one-center K_a).

VI. CONSTRUCTION ALGORITHM

The following workflow constructs Eq. (40).

A. Smooth orbital and projector data

For every orbital $(i, \mathbf{k})$, compute the smooth grid values $\tilde{\varphi}_i^{\mathbf{k}}(\mathbf{r}_g)$ and PAW projector overlaps $P_{i, a\alpha}^{\mathbf{k}}$. This is the only orbital-dependent input needed for the final THC factor $X_{\Lambda i}^{\mathbf{k}}$.

B. Global ISDF on smooth pseudo pair densities

For each transferred momentum $\mathbf{q}$, construct or sample the smooth pseudo pair-density matrix

$$\tilde{\rho}^{\mathbf{q}}(\mathbf{k}ij, \mathbf{r}_g) = \tilde{\varphi}_i^{\mathbf{k}-\mathbf{q}*}(\mathbf{r}_g) \tilde{\varphi}_j^{\mathbf{k}}(\mathbf{r}_g). \quad (46)$$

Select interpolation points $\{\mathbf{r}_\mu\}$ using pivoted Cholesky on $\tilde{S}^{\mathbf{q}} = (\tilde{\rho}^{\mathbf{q}})^\dagger \tilde{\rho}^{\mathbf{q}}$, randomized QR with column pivoting, or a rank-revealing interpolative decomposition. Then solve Eq. (10), using $\tilde{\rho}^{\mathbf{q}}$, to obtain the smooth global auxiliary functions $\zeta_\mu^{\mathbf{q}}$. As in the momentum-dependent ISDF construction, a practical simplification is to choose the points at $\mathbf{q} = 0$ and reuse them for other $\mathbf{q}$ blocks, while keeping $\zeta_\mu^{\mathbf{q}}$ momentum dependent.

C. Local ISDF in PAW channel space

For each chemical species, construct a local matrix of channel-product functions on radial/angular grids. A useful combined training set is

$$\mathcal{B}_{a, \alpha\beta}(\mathbf{s}) = \{\phi_{a\alpha}^*(\mathbf{s}) \phi_{a\beta}(\mathbf{s}), \tilde{\phi}_{a\alpha}^*(\mathbf{s}) \tilde{\phi}_{a\beta}(\mathbf{s}), \hat{Q}_{a, \alpha\beta}(\mathbf{s}), \Delta Q_{a, \alpha\beta}(\mathbf{s})\}. \quad (47)$$

Apply a local ID or pivoted Cholesky procedure to this finite channel-product set to obtain $U_{a, \lambda\alpha}$. Then solve

the local least-squares problem

$$\min_{\{\eta_{a\lambda}\}} \sum_{\alpha\beta} \left\| \hat{Q}_{a,\alpha\beta} - \sum_{\lambda} U_{a,\lambda\alpha}^* U_{a,\lambda\beta} \eta_{a\lambda} \right\|_M^2, \quad (48)$$

where M is either the local L^2 metric or a local Coulomb metric. The Coulomb metric is preferred when the goal is direct control of ERI errors.

D. One-center residual fit

Compute the exact one-center residual tensor ΔC_a from Eq. (26) using radial Coulomb integrals and Gaunt coefficients. Fit K_a from

$$\min_{K_a} \left\| \Delta C_{a,\alpha\beta,\gamma\delta} - \sum_{\lambda\xi} U_{a,\lambda\alpha}^* U_{a,\lambda\beta} K_{a,\lambda\xi} U_{a,\xi\gamma}^* U_{a,\xi\delta} \right\|_M^2. \quad (49)$$

Since this fit is atom-local and species-local, it can be performed to high accuracy and cached with the PAW data set. In the full-rank local product limit, Eq. (49) is exact.

E. Kernel assembly

Assemble $V_{\Lambda\Sigma}^{\mathbf{q},C}$ from Eq. (36). The blocks are

$$V_{\mu\nu}^{\mathbf{q},C} = (\zeta_{\mu}^{-\mathbf{q}} | \zeta_{\nu}^{\mathbf{q}}), \quad (50a)$$

$$V_{\mu,a\lambda}^{\mathbf{q},C} = (\zeta_{\mu}^{-\mathbf{q}} | \eta_{a\lambda}^{\mathbf{q}}), \quad (50b)$$

$$V_{a\lambda,b\xi}^{\mathbf{q},C} = (\eta_{a\lambda}^{-\mathbf{q}} | \eta_{b\xi}^{\mathbf{q}}). \quad (50c)$$

Then add $K_{a,\lambda\xi}$ only to same-atom local-local blocks according to Eq. (41). The final output is the pair $(X, \mathcal{V})$,

which can be consumed by any THC-based post-DFT method.

VII. RELATION TO DIRECT ALL-ELECTRON ISDF

A formally simpler but less efficient alternative is to reconstruct the all-electron orbital values at the ISDF interpolation points,

$$X_{\mu i}^{\mathbf{k},\text{AE}} = \varphi_i^{\mathbf{k}}(\mathbf{r}_{\mu}) = \tilde{\varphi}_i^{\mathbf{k}}(\mathbf{r}_{\mu}) + \sum_{a\alpha} \Delta\phi_{a\alpha}(\mathbf{r}_{\mu} - \boldsymbol{\tau}_a) P_{i,a\alpha}^{\mathbf{k}}, \quad (51a)$$

$$\Delta\phi_{a\alpha}(\mathbf{s}) = \phi_{a\alpha}(\mathbf{s}) - \tilde{\phi}_{a\alpha}(\mathbf{s}). \quad (51b)$$

One could then fit the all-electron pair density directly,

$$\rho_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) \approx \sum_{\mu} X_{\mu i}^{\mathbf{k}_i,\text{AE}*} X_{\mu j}^{\mathbf{k}_j,\text{AE}} \zeta_{\mu}^{\mathbf{k}_j - \mathbf{k}_i, \text{AE}}(\mathbf{r}), \quad (52)$$

and form a THC kernel from ζ^{AE} . This direct all-electron collocation approach is conceptually straightforward, but it requires the interpolation points and auxiliary functions to resolve core-region oscillations and cusps. The PAW-ISDF formulation avoids that cost by keeping the global grid smooth and representing core-region physics through compact atom-local features.

VIII. ERROR ANALYSIS

Let

$$\epsilon_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) = \bar{\rho}_{ij}^{\mathbf{k}_i\mathbf{k}_j}(\mathbf{r}) - \sum_{\Lambda} X_{\Lambda i}^{\mathbf{k}_i*} X_{\Lambda j}^{\mathbf{k}_j} \zeta_{\Lambda}^{\mathbf{k}_j - \mathbf{k}_i}(\mathbf{r}) \quad (53)$$

be the compensated pair-density interpolation error, and let δK_a be the local residual-fit error in Eq. (38). Define the Coulomb norm

$$\|f\|_C^2 = (f|f). \quad (54)$$

Then the smooth compensated contribution to the ERI error satisfies the bound

$$|\delta V_{ij,kl}^C| \leq \|\epsilon_{ij}^{\mathbf{k}_i\mathbf{k}_j}\|_C \|\bar{\rho}_{kl}^{\mathbf{k}_k\mathbf{k}_l}\|_C + \|\bar{\rho}_{ij}^{\mathbf{k}_i\mathbf{k}_j}\|_C \|\epsilon_{kl}^{\mathbf{k}_k\mathbf{k}_l}\|_C + \|\epsilon_{ij}^{\mathbf{k}_i\mathbf{k}_j}\|_C \|\epsilon_{kl}^{\mathbf{k}_k\mathbf{k}_l}\|_C. \quad (55)$$

The local correction contributes

$$\delta V_{ij,kl}^{\text{loc}} = \sum_a \sum_{\alpha\beta\gamma\delta} P_{i,a\alpha}^{\mathbf{k}_i*} P_{j,a\beta}^{\mathbf{k}_j} \delta C_{a,\alpha\beta,\gamma\delta} P_{k,a\gamma}^{\mathbf{k}_k*} P_{l,a\delta}^{\mathbf{k}_l}. \quad (56)$$

This separates the total error into three controllable pieces: global ISDF truncation of the pseudo pair densities, local ISDF truncation of the compensation charges,

and local fitting error of the all-electron residual tensor. The complete method is exact in the simultaneous limit of complete PAW partial waves, complete compensation multipoles, full-rank global ISDF, and full-rank local channel ISDF.

Equation (55) also suggests a practical fitting criterion. Least squares in the ordinary grid metric controls point-

wise or L^2 pair-density errors, but Coulomb-metric fitting controls the ERI error more directly. A robust implementation can therefore use an inexpensive L^2 or pivoted-Cholesky selection step followed by Coulomb-metric refinement of ζ_A^q , $\eta_{a\lambda}$, and K_a .

IX. NUMERICAL DEMONSTRATION

We demonstrate that the PAW-ISDF-THC factorization converges systematically to the exact electron repulsion integrals as the ISDF auxiliary basis is enlarged, and that the all-electron augmentation preserves this convergence. All calculations use the implementation in the CoQuif code, taking Quantum ESPRESSO mean-field data through pw2coqui. The reference system is LiH in the zinc-blende-derived rocksalt structure on a $2 \times 2 \times 2$ $\mathbf{k}$ -mesh (three irreducible $\mathbf{k}$ -points, sixteen bands), treated with three pseudization families that share the same valence: norm-conserving (NCPV), ultrasoft (USPP), and projector-augmented-wave (PAW). This lets us isolate the effect of the augmentation algebra while holding the smooth ISDF problem fixed.

A. Rank convergence of the compressed integrals

Equation (55) bounds the ERI error in the Coulomb metric by the interpolation error of the compensated smooth pair densities, controlled by the number of global interpolation points N_μ . Figure 1 shows the relative Frobenius error of the Hartree and exchange matrix elements against the direct reference as the ISDF threshold is tightened from 3×10^{-2} to 10^{-6} , sweeping N_μ from roughly 120 to 430. The error falls smoothly and nearly as a power law in N_μ , spanning about two and a half orders of magnitude for all three pseudization families. Two features are notable. First, the ultrasoft and PAW curves are essentially superimposed: the local augmentation rows in the composite auxiliary index converge at the same rate as the smooth problem and do not introduce a slower-decaying error channel. Second, the norm-conserving curve is shifted to smaller N_μ —it reaches a given accuracy with fewer interpolation points—because its pseudo pair densities require no compensation charges, whereas the ultrasoft and PAW cases add a modest amount of auxiliary content to represent the augmentation. This is precisely the behavior anticipated by Eq. (55): the augmented factorization inherits the controlled, tunable convergence of the underlying momentum-resolved ISDF, with a small and systematically convergent augmentation overhead.

B. RPA correlation energy of silicon

The Frobenius-norm test above establishes that the compressed integrals reproduce the reference all-electron

ERIs. We now show that this accuracy carries through to a physical observable evaluated entirely within the factorized representation, and that the all-electron PAW result is consistent with norm-conserving references. We compute the random-phase-approximation (RPA) correlation energy of bulk silicon (diamond structure, $a = 10.244$ Bohr) on PBE orbitals, using the PAW-ISDF-THC integrals throughout the imaginary-axis RPA, with the THC threshold fixed at 10^{-5} and the auxiliary cutoff at $1.2 \times$ the orbital cutoff ($E_{\text{cut}}^{\text{orb}} = 150$ Ry). Four pseudization families sharing the same four-electron Si valence are compared: two norm-conserving constructions (ONCV and ccECP), an ultrasoft potential, and an $l = 2$ kjpaw PAW set.

The RPA correlation energy converges along two axes that must be extrapolated independently: the number of unoccupied bands N_b entering the response function, and the density of the $\mathbf{k}$ -point mesh $N_k = n^3$. Figure 2(a) shows the band convergence at the densest mesh studied, $6 \times 6 \times 6$. For every pseudization family the correlation energy is accurately linear in $1/N_b$ for $N_b \gtrsim 100$, so a two-parameter fit over $N_b \in [100, 500]$ yields a controlled complete-basis ($N_b \rightarrow \infty$) limit (stars). The PAW and ultrasoft curves run parallel to the norm-conserving ones with no anomalous slowly decaying component: the augmentation algebra does not degrade the band convergence. Figure 2(b) then extrapolates the $\mathbf{k}$ -mesh. The band-converged values are linear in $1/N_k$ once the mesh is sufficiently dense; the $3 \times 3 \times 3$ points (shown) still lie above the asymptote, so the thermodynamic-limit fit uses only $n \geq 4$.

The doubly-extrapolated ($N_b, N_k \rightarrow \infty$) RPA correlation energies are -0.460 Ha (ONCV) and -0.455 Ha (ccECP) for the two norm-conserving sets, -0.448 Ha for the kjpaw PAW set, and -0.446 Ha for the ultrasoft potential. The PAW value differs from the norm-conserving references by about 1–3% (1.4% from ccECP, 2.5% from ONCV), comparable to the $\sim 1\%$ spread between the two norm-conserving potentials themselves. This is the expected level of agreement between distinct pseudization families for an absolute correlation energy: the PAW-ISDF-THC factorization introduces no error beyond the intrinsic scatter of the underlying potentials, confirming that the all-electron augmentation is both systematically convergent and physically faithful when driven through a full many-body calculation.

X. COMPUTATIONAL SCALING AND STORAGE

Let N_{orb} be the number of orbitals per $\mathbf{k}$ point, N_k the number of $\mathbf{k}$ points, N_μ the number of global interpolation points, and

$$N_A = \sum_a n_a \quad (57)$$

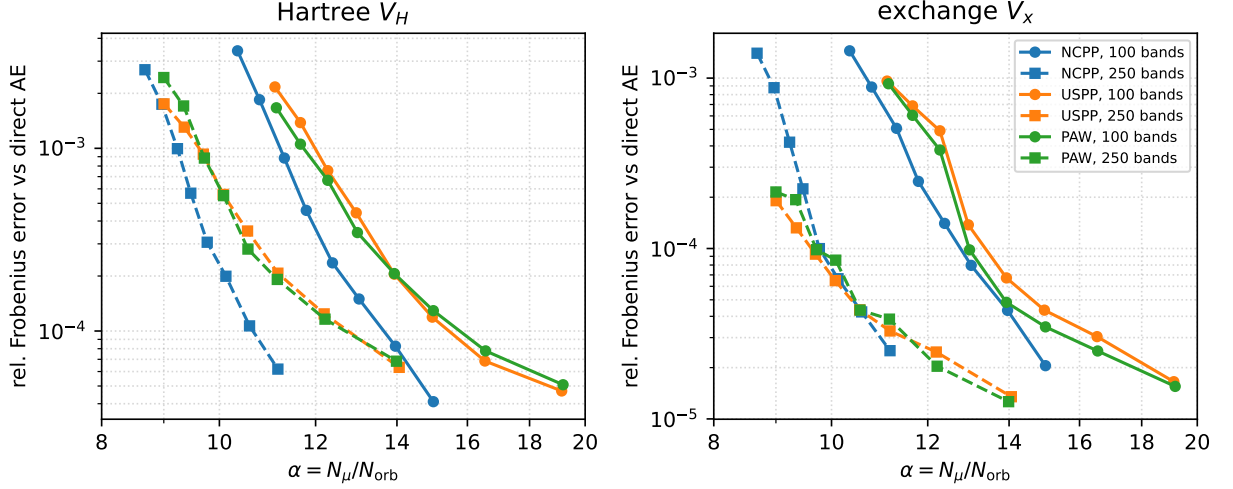


FIG. 1. Relative Frobenius error of the tensor-hypercontracted Hartree (left) and exchange (right) matrix elements versus the number of ISDF interpolation points N_μ , for LiH on a $2 \times 2 \times 2$ mesh with norm-conserving (NCPP), ultrasoft (USPP), and projector-augmented-wave (PAW) pseudization. Tightening the ISDF threshold from 3×10^{-2} to 10^{-6} drives the error down by about two and a half orders of magnitude. PAW converges at the same rate as the ultrasoft case (the two curves overlap), while the norm-conserving case reaches a given accuracy at smaller N_μ because it carries no compensation charges.

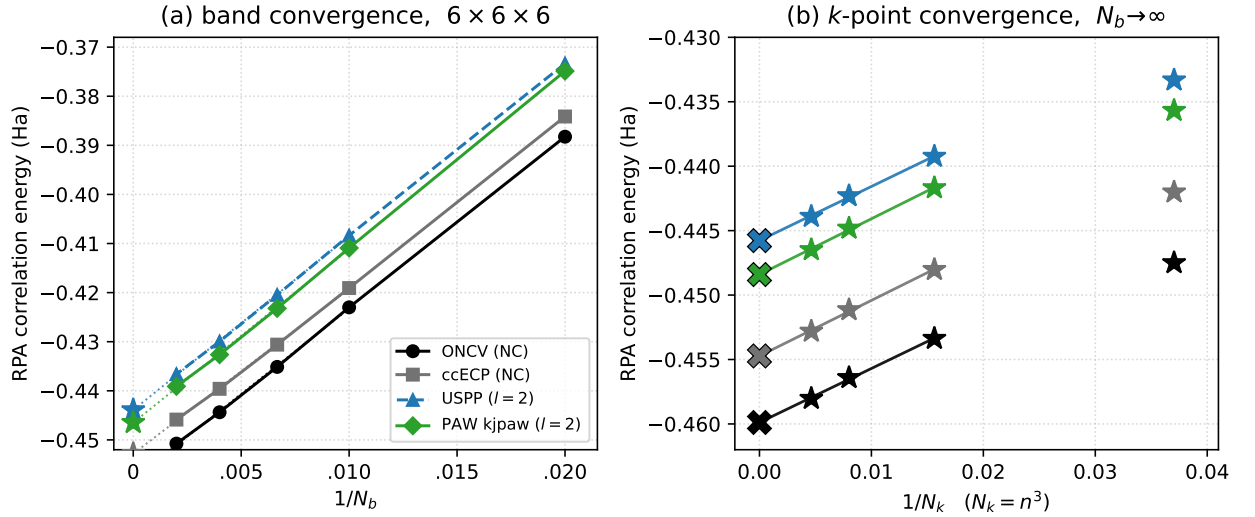


FIG. 2. RPA correlation energy of bulk silicon computed with the PAW-ISDF-THC integrals, for two norm-conserving (ONCV, ccECP), one ultrasoft, and one $l = 2$ **kjpaw** PAW pseudization, all sharing the four-electron valence. (a) Basis-set convergence at the densest $6 \times 6 \times 6$ mesh: the correlation energy is linear in $1/N_b$ for $N_b \gtrsim 100$; the dotted lines are linear fits over $N_b \in [100, 500]$ and the stars mark the complete-basis ($N_b \rightarrow \infty$) limit. (b) **k**-point convergence of the band-extrapolated values versus $1/N_k$ ($N_k = n^3$): stars are shown for $n \geq 3$, the solid lines are linear fits over the asymptotic regime $n \geq 4$, and the crosses mark the thermodynamic ($N_b, N_k \rightarrow \infty$) limit. The all-electron PAW result tracks the norm-conserving references to within the $\sim 1\%$ scatter of the norm-conserving potentials among themselves.

the total number of atom-local augmentation rows per unit cell. The total PAW-ISDF THC rank is

$$N_\Lambda = N_\mu + N_A. \quad (58)$$

The factor storage is

$$\text{storage}(X) = O(N_k N_{\text{orb}} N_\Lambda), \quad \text{storage}(\mathcal{V}) = O(N_q N_\Lambda^2), \quad (59)$$

with possible reductions from time-reversal, point-group, and momentum symmetries. When $N_\mu \sim \alpha N_{\text{orb}}$ and $N_A \sim O(N_{\text{orb}})$, downstream THC contractions retain

the cubic-in-system-size and linear-in- N_k structure characteristic of momentum-resolved ISDF formulations, but with a modest PAW augmentation overhead.

The dominant construction costs are: (i) building or sampling smooth pair densities, (ii) selecting interpolation points and solving Eq. (10), (iii) computing PAW projector features, (iv) assembling the Coulomb matrix in the composite auxiliary basis, and (v) fitting the species-local tensors η_a and K_a . Steps (iii) and (v) are low-dimensional compared with global pair-density operations. Step (iv) can be accelerated by FFTs for smooth components and by analytic or multipole representations for local components.

XI. DISCUSSION

The central result of this work is that interpolative separable density fitting extends naturally from norm-conserving pseudopotentials to the augmented families. Both ultrasoft and PAW pair densities are reduced to a compensated smooth density plus a finite one-center correction; the smooth part is interpolated by ordinary ISDF on a moderate uniform grid, while the augmentation enters through a small set of atom-centered auxiliary rows, with PAW adding a single same-atom block K_a that the ultrasoft scheme omits (Sec. V A). Three properties follow. First, the global ISDF problem stays smooth, because the interpolation points represent $\tilde{\varphi}_i^* \tilde{\varphi}_j$ rather than the rapidly oscillatory all-electron products. Second, for PAW the all-electron core-region information is retained through the partial waves and the one-center matrix K_a . Third, the output has exactly the tensor-hypercontracted structure—and the cubic scaling and linear $\mathbf{k}$ -point dependence—of the norm-conserving case, so existing THC machinery for RPA, GW , and quantum embedding can be reused with only a modestly enlarged auxiliary dimension. The numerical results bear this out: the compressed integrals converge as a power law to the reference ultrasoft and PAW integrals (Fig. 1), and the silicon RPA correlation energy evaluated entirely within the factorization agrees, for both ultrasoft and PAW, with norm-conserving references to within the intrinsic spread of the pseudopotentials themselves (Fig. 2).

The central numerical question is the choice of local rank. A minimal implementation can take the local factorization in Eq. (38) to be full rank in the PAW channel-product space. Since the number of PAW channels per atom is small, this may already be affordable and avoids local fitting error. A more aggressive implementation can compress the channel products by a local ISDF or Cholesky procedure. This becomes useful for semicore-rich PAW data sets, high angular momentum channels, spin-orbit projectors, or multi-projector setups.

The compensation-charge construction must be consistent with the PAW data set. If $\hat{Q}_{a,\alpha\beta}$ does not reproduce the multipoles of $\Delta Q_{a,\alpha\beta}$, Eq. (24) is not a clean separation into smooth and one-center terms. In that case, spu-

rious inter-sphere terms would be folded into the global ISDF error and convergence with respect to N_μ could become poor. Therefore the PAW compensation multipole order should be converged before the ISDF rank is tuned.

A closely related prerequisite is the completeness of the PAW partial-wave and projector set. It is well established that PAW data sets constructed for ground-state density-functional theory—where the partial waves need only reproduce all-electron scattering near the occupied levels—are frequently inadequate for GW and RPA, which sample the unoccupied manifold to much higher energies; accurate excited-state calculations therefore require additional projectors at higher reference energies and, where necessary, higher angular momenta. This is precisely the rationale for the dedicated “ GW ” PAW data sets introduced by the VASP group [15, 16]. Our factorization inherits this requirement directly: the construction reproduces the all-electron integrals of whatever data set it is given, so any partial-wave incompleteness of that data set propagates into the integrals, and the one-center correction in Eq. (25) is quartic in the projector overlaps $P_{i,a\alpha}$. In production one should therefore use excited-state-quality (“ GW ”) PAW data, or otherwise verify partial-wave completeness over the energy window sampled by the method.

The Coulomb singularity requires the same care as in ordinary periodic ERIs. For $\mathbf{q} \neq 0$, Eq. (37) is well-defined. For $\mathbf{q} = 0$, the $\mathbf{G} = 0$ term is treated by charge neutrality, by a neutralizing background, by Ewald correction, or by the convention used by the target many-body method. The PAW compensation charges should be included in the neutralized pair density before this step.

Frozen-core PAW introduces additional issues if one wants ERIs involving core states. The valence-valence ERI in Eq. (40) is the most common target for post-DFT valence Hamiltonians. Core-valence and core-core exchange or correlation terms can be added as additional atom-local tensors, analogous to K_a , because frozen core orbitals are atom-centered and species-local. If explicit core orbitals are included in the single-particle basis, they can be appended as additional local rows in X rather than represented on the global smooth grid.

Spinor and noncollinear calculations can also be incorporated. The orbital index i then carries spinor components, the projector overlaps become spinor projector overlaps, and the pair density includes spin contractions appropriate to the target interaction. For the spin-independent Coulomb operator, the spatial PAW-ISDF kernel $\mathcal{V}^{\mathbf{q}}$ is unchanged; only the construction of the pair factors $X_{\Lambda_i}^{k*} X_{\Lambda_j}^{k'}$ is modified.

XII. CONCLUSIONS

We have presented an extension of momentum-resolved ISDF and THC to ultrasoft and projector-augmented-wave pseudopotentials. The formulation starts from the

exact electrostatic partition of an augmented pair density into a compensated smooth density and atom-local residual corrections. The compensated smooth density is represented by a composite ISDF auxiliary basis containing ordinary smooth interpolation rows and atom-centered augmentation rows. For PAW, the one-center all-electron residual is factored in the same local channel basis and added to the same-atom block of the projected Coulomb matrix; for ultrasoft pseudopotentials this block is absent and the same construction applies unchanged, with norm-conserving pseudopotentials recovered when the augmentation rows are dropped. The final expression,

$$V_{ijkl}^{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_k \mathbf{k}_l} \approx \sum_{\Lambda \Sigma} X_{\Lambda i}^{\mathbf{k}_i *} X_{\Lambda j}^{\mathbf{k}_j} V_{\Lambda \Sigma}^{\mathbf{q}} X_{\Sigma k}^{\mathbf{k}_k *} X_{\Sigma l}^{\mathbf{k}_l}, \quad (60)$$

retains the standard THC separability of orbital and momentum indices while incorporating the augmentation algebra of both schemes. This makes the approach a natural route to ultrasoft- and all-electron-quality hybrid, RPA, *GW*, embedding, and wave-function calculations starting from the augmented mean-field orbitals produced by standard plane-wave codes.

Appendix A: Compact derivation of the PAW Coulomb partition

Let

$$\Delta \rho_{ij}^a = \rho_{ij}^{1,a} - \tilde{\rho}_{ij}^{1,a}, \quad \delta \rho_{ij}^a = \Delta \rho_{ij}^a - \hat{\rho}_{ij}^a. \quad (A1)$$

Then

$$\rho_{ij} = \bar{\rho}_{ij} + \sum_a \delta \rho_{ij}^a. \quad (A2)$$

Because each $\delta \rho_{ij}^a$ has zero multipole moments through the compensation order, its electrostatic potential vanishes outside the augmentation sphere. For non-overlapping augmentation spheres,

$$(\delta \rho_{ij}^a | \bar{\rho}_{kl}) + (\bar{\rho}_{ij} | \delta \rho_{kl}^a) + \sum_{b \neq a} (\delta \rho_{ij}^a | \delta \rho_{kl}^b) = 0 \quad (A3)$$

up to the chosen multipole completeness. Therefore

$$(\rho_{ij} | \rho_{kl}) = (\bar{\rho}_{ij} | \bar{\rho}_{kl}) + \sum_a (\delta \rho_{ij}^a | \delta \rho_{kl}^a). \quad (A4)$$

Using $\delta \rho^a = \rho^{1,a} - (\tilde{\rho}^{1,a} + \hat{\rho}^a)$, Eq. (A4) is equivalent to Eq. (24). The finite-dimensional tensor ΔC_a is obtained by inserting the partial-wave expansions of the one-center pair densities.

Appendix B: Least-squares solution for local channel THC

Let $I = (\alpha\beta)$ and λ label channel products and local interpolation rows. Define

$$B_{\lambda I}^a = U_{a,\lambda\alpha}^* U_{a,\lambda\beta}. \quad (B1)$$

The compensation-charge fit in Eq. (48) is the matrix problem

$$\hat{Q}_I^a(\mathbf{s}) \approx \sum_{\lambda} B_{\lambda I}^a \eta_{a\lambda}(\mathbf{s}). \quad (B2)$$

For each grid point or radial basis coefficient m , this becomes

$$\mathbf{q}_m^a \approx (B^a)^T \boldsymbol{\eta}_m^a. \quad (B3)$$

The local residual fit is similarly

$$\Delta C_a \approx (B^a)^\dagger K_a B^a, \quad (B4)$$

where ΔC_a is viewed as a matrix in the compound channel-product index I . If B^a has full row rank, the least-squares solution is

$$K_a = [(B^a)^\dagger]^\dagger \Delta C_a (B^a)^\dagger, \quad (B5)$$

with $+$ denoting the Moore-Penrose pseudoinverse, optionally evaluated in a metric-weighted inner product.

ACKNOWLEDGMENTS

Numerical validation against direct PAW Coulomb matrix elements (Fig. 1) and against a fully converged RPA observable (Fig. 2) is reported in Sec. IX; extension to broader benchmark sets and production many-body workflows is ongoing.

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